

3.10-9 Filtro de Média Móvel

N pontos

$$h[n] = \frac{1}{N} (u[n] - u[n-N])$$

a) Equação diferença $\sum_{k=0}^{N-1} \delta[n-k]$ pulso retangular

p/ uma entrada $x[n]$, a saída

será $y[n]$ $\therefore y[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[n-k]$

(b) python e (c)

3.10-6 $h[n] = \left(\frac{1}{2}\right)^{|n|}$

a) este sistema é causal?

solução causal $h[n] = 0$ p/ $n < 0$

Como $h[-1] \neq 0 \therefore$ sistema NÃO causal

b) O sistema é BIBO?

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

$$\sum_{n=-\infty}^{\infty} \left|\left(\frac{1}{2}\right)^{|n|}\right| = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^{|n|} = \sum_{n=-\infty}^{-1} \left(\frac{1}{2}\right)^{-n} + \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \sum_{n=-\infty}^{-1} 2^n + \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$$

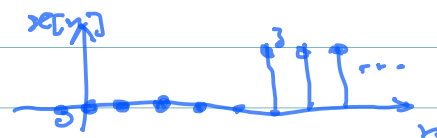
$$\begin{aligned} \text{PG} \left[\sum_{k=m}^n q^k = \frac{q^m - q^{n+1}}{1-q} \right] &= \frac{2^{-\infty} - 2^0}{1-2} + \frac{\left(\frac{1}{2}\right)^0 - \left(\frac{1}{2}\right)^{\infty}}{1-\frac{1}{2}} \\ &= \frac{\left(\frac{1}{2}\right)^{\infty} - 1}{-1} + \frac{1-0}{\frac{1}{2}} \\ &= 1 + 2 \end{aligned}$$

$$\sum |h[n]| = 3$$

Logo sistema é BIBO estável

$$c) E = ? \quad P = ? \quad x[n] = 3u[n-5]$$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \infty$$



$$E = \sum_{n=0}^{\infty} 3^2 = \infty$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2 = \lim_{N \rightarrow \infty} \sum_{n=-N}^N 9$$

$$P = 0 + \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=5}^N 9 = 9 \lim_{N \rightarrow \infty} \frac{1}{2N+1} (N-5+1) = 9 \frac{N}{2N} = \frac{9}{2}$$

Trata-se de um sinal de potência

d) Determine a resposta para o estado nulo. $y[10] = ?$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k]$$

$$h[k] = \left(\frac{1}{2}\right)^{|k|}$$

$$x[n-k] = 3u[n-k-5] \quad y[10] = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{|k|} 3u[10-k-5]$$

$$y[10] = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{|k|} 3u[5-k] = 3 \left\{ \sum_{k=-\infty}^{-1} \left(\frac{1}{2}\right)^{-k} + \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \right\}$$

$$y[10] = 3 \left\{ \sum_{k=-\infty}^{-1} 2^k + \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \right\} = 3 \left\{ \frac{2^{\infty} - 2^0}{1-2} + \frac{\left(\frac{1}{2}\right)^0 - \left(\frac{1}{2}\right)^{\infty}}{1 - \frac{1}{2}} \right\}$$

$$y[10] = 3 \left\{ 1 + 2 \left(\frac{64-1}{64} \right) \right\} = 3 \left(\frac{32+63}{32} \right)$$

$$y[10] = \frac{285}{32}$$

3.10-5

$$h[n] = \delta[n] + \left(\frac{1}{3}\right)^n u[n-1]$$

a) Este sistema é estável? é causal?

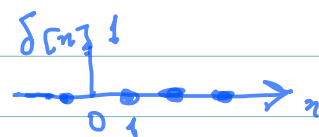
Solução

$$h[n] = \left(\frac{1}{3}\right)^n u[n]$$

Sistema estável

$$\text{pois } \sum |h[n]| < \infty$$

Sistema causal pois $h[n] = 0$ p/ $n < 0$



b) Python código

$$n = \text{np.arange}(-10, 10)$$

$$x = u[n-3] - u[n+3]$$

$$\text{plt.stem}(x, n)$$

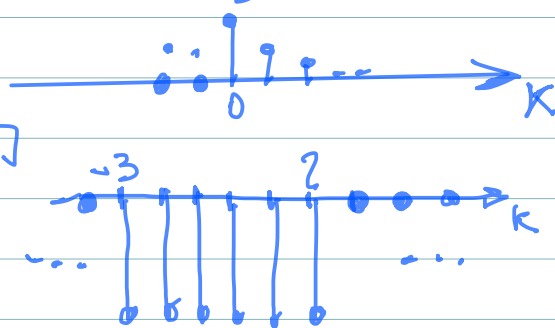
c) Determine $y[n]$? Resposta ao Gafado Nulo

$$y[n] = x[n] * h[n]$$

$$h[k] = \left(\frac{1}{3}\right)^k u[k]$$

$$x[k] \neq 0 \text{ p/ } -3 \leq k \leq 2$$

$x[k]$



$$h[k] \neq 0 \text{ p/ } k \geq 0$$

$$h[n-k] \neq 0 \text{ p/ } n-k \geq 0$$

$$\Rightarrow y[n] = ? \quad \boxed{k \leq n}$$

$$i) \quad n < -3 \Rightarrow y[n] = 0$$

$$ii) \quad -3 \leq k \leq 2 \Rightarrow y[n] = \sum_{k=-3}^n (-1) \left(\frac{1}{3}\right)^{n-k} = -\left(\frac{1}{3}\right)^n \sum_{k=-3}^n 3^k$$

$$y[n] = -\left(\frac{1}{3}\right)^n \frac{3^{-3} - 3^{n+1}}{1-3} = \frac{1}{2} \cdot 3^{-n} (3^{-3} - 3^{n+1})$$

$$y[n] = \frac{1}{2} (3^{-(n+3)} - 3)$$

$$iii) \text{ p/ } k > 2 \quad y[n] = \sum_{k=-3}^2 (-1) \left(\frac{1}{3}\right)^{n-k} = -\left(\frac{1}{3}\right)^n \sum_{k=-3}^2 3^k$$

$$y[n] = -\left(\frac{1}{3}\right)^n \frac{3^3 - 3^3}{1-3} = \frac{1}{2} 3^{-n} (3^{-3} - 3^3) = \frac{3^{-n}}{2} \left(\frac{1-27.27}{27} \right)$$

$$y[n] = -\frac{728}{54} \cdot 3^{-n} = -\frac{364}{27} \cdot 3^{-n}$$

3.9-8 $(E^2 - E + 0,16)y[n] = Ex[n]$; $x[n] = \cos(\frac{\pi}{2}n + \frac{\pi}{3})u[n]$

$$y[-1] = y[-2] = 0$$

$$y[0] = ? \quad y[1] = ?$$

Iterativamente

solução

$$y[n+2] - y[n+1] + 0,16y[n] = x[n+1] \quad \text{eq. 1}$$

$$p/n = -2$$

$$y[0] - y[-1] + 0,16y[-2] = x[-1]$$

$$y[0] - 0 + 0 = 0 \Rightarrow y[0] = 0$$

$$p/n = -1$$

$$y[1] - y[0] + 0,16y[-1] = x[0]$$

$$y[1] - 0 + 0 = \cos(\frac{\pi}{3}) = \frac{1}{2}$$

$$y[1] = \frac{1}{2}$$

Forma de $y[n]$

$$y[n] = K \cos(\frac{\pi}{2}n + \frac{\pi}{3} + \phi)$$

Na equação eq. 1

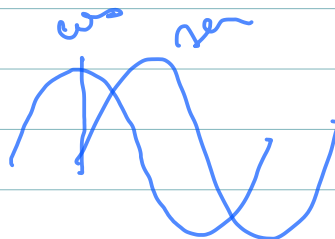
$$K \cos(\frac{\pi}{2}(n+2) + \frac{\pi}{3} + \phi) - K \cos(\frac{\pi}{2}(n+1) + \frac{\pi}{3} + \phi) + 0,16K \cos(\frac{\pi}{2}n + \frac{\pi}{3} + \phi) = \cos(\frac{\pi}{2}(n+1) + \frac{\pi}{3})$$

$$-K \cos(\frac{\pi}{2}n + \frac{\pi}{3} + \phi) - K \sin(\frac{\pi}{2}n + \frac{\pi}{3} + \phi) + 0,16K \cos(\frac{\pi}{2}n + \frac{\pi}{3} + \phi) = \sin(\frac{\pi}{2}n + \frac{\pi}{3})$$

$$-0,84K \cos(\frac{\pi}{2}n + \frac{\pi}{3} + \phi) - K \sin(\frac{\pi}{2}n + \frac{\pi}{3} + \phi) = \sin(\frac{\pi}{2}n + \frac{\pi}{3})$$

$$K = 0,765$$

$$\begin{array}{r} 100 \\ - 0,16 \\ \hline 84 \end{array}$$



$$\phi = -2,44 \text{ rad}$$

3.9-5 (a) $\sum_{k=0}^n k = ?$

$$y[n] = \sum_{k=0}^n k$$

$$y[n+1] = \sum_{k=0}^n k + n+1$$

$$y[n+1] - y[n] = n+1 \quad *$$

E: Operador Avanço

$$(E - 1)y[n] = n+1$$

Root (autovvalor) $\lambda = 1$ - solução homogênea (característica)

$$y_h[n] = K_1 1^n = K_1$$

$$y_p[n] = c_0 + c_1 n \quad (\text{Tabela Livro})$$

n não permitido (já característico)

$$y_p[n] = c_1 n + c_2 n^2$$

Substituindo em (*)

$$c_1(n+1) + c_2(n+1)^2 - c_1 n - c_2 n^2 = n+1$$

$$\cancel{c_1 n} + \cancel{c_1} + \cancel{c_2 n^2} + 2c_2 n + \cancel{c_2} - \cancel{c_1 n} - \cancel{c_2 n^2} = n+1$$

$$\begin{cases} c_1 + c_2 = 1 \\ 2c_2 = 1 \end{cases}$$

$$\Rightarrow c_2 = \frac{1}{2} \Rightarrow c_1 = \frac{1}{2}$$

$$\therefore y[n] = K_1 + \frac{1}{2}n + \frac{1}{2}n^2 \quad y[0] = 0$$

$K_1 = 0$

$$\therefore y[n] = \sum_{k=0}^n K_1 = \frac{n(n+1)}{2}$$

W

(b) $y[n] = \sum_{k=0}^n k^3$

$$y[n+1] - y[n] = (n+1)^3$$

$$y_h[n] = K_1$$

$$y_p[n] = c_1 n + c_2 n^2 + c_3 n^3 + c_4 n^4$$

$$C_1 = 0, \quad C_2 = 1/4, \quad C_3 = 1/2 \quad \text{e} \quad C_4 = 1/4$$

$$y[n] = K_1 + \frac{1}{4}n^2 + \frac{1}{2}n^3 + \frac{1}{4}n^4 \quad y[n] = 0 \Rightarrow K_1 = 0$$

$$y[n] = \frac{n^2(n+1)^2}{4}$$

Exercícios cap 3 (Resolvidos)

Ex 1 1-1, 1-2, 3-1, 3-2, 3-3, 3-4,

4-13, 5-2, 5-4, 6-7, 8-1, 4-6, 4-9,

1-3, 1-4, 4-5, 8-14, 8-19,

2-2, 6-5, 7-1, 11-8, 10-6, 10-5, 9-8, 9-5

3.2-2 Potência de $x[n]$, período, e P_x

a) $-x[n]$ $P = \frac{E}{\Delta n}$ e $E \propto |x[n]|^2$

$$|-x[n]|^2 = |x[n]|^2 \quad \therefore P_a = P_x$$

b) $x[-n]$ $P_b = P_x$

c) $x[n-m]$ A energia de um sinal é a mesma, se ele for deslocado.
 $\therefore P_c = P_x$

d) $cx[n]$ $E = c^2 E_x \Rightarrow P = c^2 P_x$

e) $x[m-n]$ $E = E_x \Rightarrow P = P_x$

3.6-5 Fibonacci

seq: $\{0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots\}$

(a) Eşya için diferensiyel

$$f[n] = f[n-2] + f[n-1]$$

$$f[1] = 0$$

$f[2] = 1$ Não há entrada $\sigma[7] = 0$

(b) $f[n+2] - f[n+1] - f[n] = 0 \quad (\forall n \geq 0)$

Eq. characteristics, $r^2 - r - 1 = 0$

$$A = 1 + 4 = 5$$

$$r_2 = \frac{1 \pm \sqrt{5}}{2}$$

$$\rightarrow r_1 = \frac{1 + \sqrt{5}}{2} \approx 1,618$$

$$r_2 = \frac{1 - \sqrt{5}}{2} \approx -0,618$$

Sistema
Instável!

(c) $f[n] = c_1 \gamma_1^n + c_2 \gamma_2^n$

$$\int 0 = C_1 \cdot 1,618 - C_2 \cdot 0,618$$

$$\begin{cases} 1 = c_1(1,618)^2 + c_2(0,618)^2 \end{cases}$$

$$c_1 = 0,2764$$

$$C_2 = 0,7236$$

$$f_{50} = 0,2764 (1,618)^{50} + 0,7236 (0,618)^{50}$$

$$0 = r_1 C_1 + r_2 C_2 \quad \leftarrow r_1$$

$$1 = r_1^2 C_1 + r_2^2 C_2$$

$$C_1 = - \frac{r_2 C_2}{r_1}$$

(-)

$$-1 = \frac{1}{r_1 r_2 C_2 - r_2^2 C_2} \Rightarrow C_2 = \frac{-1}{r_1 r_2 - r_2^2}$$

Python

$$f_{50} = 7.788742 \text{E}+09$$

$$f_{1000} = 2,686381 \text{ e} + 208$$

7.1- $h[n] = ?$ (a) $y[n+1] + 2y[n] = x[n]$
 (b) $y[n] + 2y[n-1] = x[n]$

Solution $h[n] = A_0 \delta[n] + y_c[n] u[n]$

a) $h[n+1] + 2h[n] = \delta[n]$ $A_0 = \frac{b_N}{a_N}$
 $A_0 = \frac{1}{2}$

$r + 2 = 0 \Rightarrow r = -2$

$\therefore h[n] = \frac{1}{2} \delta[n] + c(-2)^n u[n]$

$h[-1] = 0$

$h[0]$

$h[0] + 2h[-1] = \delta[-1]$

$h[0] = 0$

Logo

$h[0] = \frac{1}{2} \delta[0] + c \cdot 1 = 0 \Rightarrow c = -\frac{1}{2}$

Finalmente, $h[n] = \frac{1}{2} \delta[n] - \frac{1}{2} (-2)^n u[n]$

b) $h[n+1] + 2h[n] = \delta[n+1]$

$A_0 = \frac{0}{2} = 0$

$r + 2 = 0 \Rightarrow r = -2$

$h[n] = c(-2)^n u[n]$

$h[-1] = 0$

$h[0] + 2h[-1] = \delta[0]$

$h[0] + 0 = 1$

$1 = c(-2)^0 \Rightarrow \boxed{c = 1}$

$\therefore h[n] = (-2)^n u[n]$